A PALAEOMAGNETIC STUDY OF JURASSIC AND CRETACEOUS SEDIMENTS

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Summary

Representative samples of Cretaceous and Jurassic arenaceous rocks and some Jurassic limestone samples were collected. The arenaceous rocks were generally either too weakly magnetized for the measurement of their remanent magnetization, or were unstable; higher intensities of magnetization were recorded in the limestones. The magnetic directions of the few reversed Jurassic limestones indicate a pole position near Omsk. X-ray analysis indicates that the detrital iron ore mineral present was goethite. It is concluded that the palaeomagnetic unsuitability of these Jurassic and Cretaceous deltaic deposits results from the weathering conditions in the source area.

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1. *Introduction.*—The collections of Jurassic and Cretaceous rocks were made in an attempt to delimit pole positions between those established for Triassic rocks by Clegg, Almond and Stubbs (**1**). Creer, Irving and Runcorn (**2**) and Runcorn (**3**), and for Tertiary rocks by Hospers (**4**), Hospers and Charlesworth (**5**) and Runcorn (**6**). The advice of Professor S. K. Runcorn, the help of Professor P. Allen in the planning of Cretaceous sampling, of Dr H. Megaw and the Crystallography Department of the Cavendish Laboratory in the X-ray study of detrital minerals, is gratefully acknowledged. Over one hundred and fifty samples were collected in the Weald, North Yorkshire, Skye, Sutherland, Northern France, and Luxemburg. Measurements were made in Cambridge and Newcastle on an astatic magnetometer in the manner detailed by Collinson, Creer, Irving and Runcorn (**7**).

2. *Cretaceous rocks.*—The arenaceous lower Cretaceous lithologies of the Wealden district (see Table I), the Upper Greensand and White Sandstone of Morvern, the Upper Greensand of Devon, and the Upper Albian and Upper Aptian of St Dizier (Marne) were collected. Creer reports a fairly strong but unstable natural remanent magnetization from the Hunstanton Red Rock. Difficulty was experienced in sampling owing to the friable nature of many of the beds. Lithological descriptions and successions are found in Edmunds (**8**) and Evans and Stubblefield (**9**). The hand specimens were generally pale coloured, grey, white or buff while the more argillaceous samples were a blue-grey in colour. They were in general well-sorted, fine or very fine sands on the Wentworth scale (see Krumbein and Pettijohn (**10**), p. 135) with a low heavy mineral yield from bromoform separations of 0·03 to 0·05 per cent, of which the iron ore minerals form over half. Separations of iron ore minerals made by an electromagnet and handpicking were X-rayed in a 9 cm cobalt-filtered camera. Long exposures were usually necessary and the few rather diffuse lines indicative of goethite suggest a very fine state of division.

The arbitrary lower limit of measurement adopted was when the top of the disk at 2·5 cm from the lower magnet failed to produce a minimum deflection of 0·5 cm using a 10 m optical lever. This is equivalent to an intensity of less

TABLE I

The Lower Cretaceous succession of the Weald, indicating the number of samples, and the number of sites from which arenaceous lithologies were collected

Formation			Thickness in feet	Samples	Sites
Gault Clay (Iron Grit at base 2"-6")			100 - 250	26	8
Lr. Greensand	Folkestone beds		5 - 250	11	6
	Sandgate beds		30 - 120		
	Hythe beds		0 - 350	—	—
	Atherfield Clay		0 - 60	—	—
Weald Clay			400 - 1400	—	—
Hastings beds	Tunbridge Wells Sand	Lingfield beds	130 - 400	—	—
		U. Tunbridge Wells Sand and Cuckfield Clay		4	2
		Grinstead Clay		2	1
		Lr. Tunbridge Wells Sand		10	4
	Wadhurst Clay		100 - 220	4	2
	Ashdown Sand		160 - 700	21	3
	Fairlight Clay		0 - 400	—	—

than $1{\cdot}7 \times 10^{-7}$ e.m.u./cm^3 for the usual value of the instrument sensitivity. On this basis the majority of the rocks collected were too weakly magnetized for measurement. Those with slightly higher intensities, between 0·2 and $0{\cdot}6 \times 10^{-6}$ e.m.u./cm^3, were found to be unstable. The error in the measurement of the magnetic directions is estimated to be less than 3°.

The only horizon which had a strong permanent magnetization, in which the intensity of magnetization ranged between 0·2 and $25{\cdot}0 \times 10^{-5}$ e.m.u./cm^3, was the Iron Grit at the base of the Gault. The Iron Grit, a red, pebbly, nodular band from 2 to 6 inches thick, is regarded by Kirkcaldy (11) as a sea floor deposit in which the iron-containing mineral (glauconite) was contemporaneously oxidized. Unaltered glauconite occurs in the non-magnetic Upper Greensand so that it is evident that the permanent magnetization of the Iron Grit is of a chemical nature. The directions of magnetization are nearly parallel to the present field; both normal and reversed directions being obtained.

3. *Jurassic rocks.*—Collections were made in Sutherland, Skye, North Yorkshire, Northern France and Luxemburg of sandstones ranging in age from Liassic to Kimmeridgian. Two marine horizons in the Yorkshire Estuarine (Deltaic) Series, the Ellerbeck and Millepore Beds, and limestones near the base of the Estuarine Series, and in the Lower Lias of Skye were also sampled. Specimens of the Oxford Clay near Peterborough were also obtained.

Dr K. M. Creer has made available results from a few specimens collected in Dorset which he considers to be unstable (from the Lias, Inferior Oolite, Fullers Earth, Forest Marble and Abbotsbury Ironstone).

The horizons sampled are indicated in Table II, and lithological and stratigraphical details can be found in Wilson (**12**), Phemister (**13**), Judd (**14**), Lee and Pringle (**15**), MacGregor (**16**), Arkell (**17**), Fox-Strangeways and Barrow (**18**), and Evans and Stubblefield (**9**).

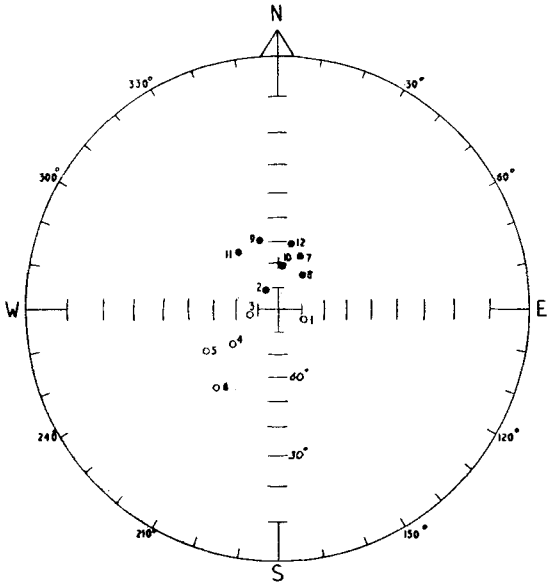
TABLE II

A correlation chart of the Lower Jurassic rocks of North Yorkshire, Skye, and Sutherland, indicating the thickness in feet of the horizons sampled, while the number in brackets represents the number of samples collected

North Yorkshire	West of Scotland (Skye)	East of Scotland (Sutherland)
Kimmeridge Clay		Loth River Shales and Sandstone
Corallian: Upper Calcareous Grit; Osmington Oolitic Series; Middle Calcareous Grit; Hambleton Oolitic Series; Lower Calcareous Grit 130′ (2)		? U. Corallian 100′ (9)
Oxford Clay; Hackness Rock; Kellaways Rock; Cornbrash	Clay zonal range uncertain	Brora Argillaceous Series 180′ (12); Brora Argillaceous Series
Great Oolite: Upper Estuarine Series 200′ (6)	Flaggy Limestone of Raasay; Estuarine Beds 270′ (7)	Estuarine Beds (base not seen) 250′ (4)
Inferior Oolite: Scarborough Grey Limestone; Middle Estuarine Series 100′ (9); Millepore Oolite 30′ (2); Lower Estuarine Series 300′ (6); Dogger	Lower Oolitic Sandstone 600′ (9); Limestone and sandy shales	faulted out
Lias: Upper 150′ (2); Middle 180′ (4); Lower	Raasay ironstone and shales; Scalpa Beds 260′ (4); Pabba Beds; Broadford Beds 500′ (5)	Lias of Dunrobin 170′ (6)

The arenaceous samples were generally grey-white or buff in colour, well-sorted, medium-grained sands, while the more argillaceous sandstones were a darker, blue-grey colour. The variable heavy mineral content, determined from a limited number of specimens, averaged about 0·05 per cent (slightly higher than in the Cretaceous rocks examined). X-ray analyses of the iron ore concentrates were again made, and the lines were suggestive of goethite in a better state of crystallinity than that found in Cretaceous rocks.

Only a few of the Yorkshire arenaceous samples (Fig. 1, 7–12) were sufficiently strongly magnetized, with intensities of magnetization between 0·5 and $1{\cdot}0 \times 10^{-6}$ e.m.u./cm^3, to be measured. The directions of magnetization obtained (Fig. 1) appeared consistent, with a direction close to the dipole field, but on remeasurement after a period of random storage of eighteen months they were rejected as unstable. An examination of the statistics calculated at the 5 per cent level of confidence (Fisher (**19**)), showed that random storage had little affected the mean, but had increased the angle of the cone of confidence (α) from 4°·7 to 6°·7. The few Scottish sandstones which could be measured were with a single exception more obviously unstable. The Oxford Clay was too weakly magnetized to be measured.



DIRECTION OF MAGNETIZATION OF JURASSIC ROCKS.

FIG. 1.

Plot of the declination and inclination of Jurassic samples (Wulff stereographic projection).

\+ *Dipole field*
● *N-seeking pole pointing downwards*
○ *N-seeking pole pointing upwards.*
Key to sample numbers:

1, 2	*limestones from the Lias*	*Skye*
3, 4, 5	*limestones from the Estuarine beds*	*Skye*
6	*sandstone from the Brora Arenaceous Series, Sutherland*	
7	*sandstone from the Lias*	*North Yorkshire*
8	*sandstone from the Lower Estuarine beds*	*North Yorkshire*
9	*sandstone from the Middle Estuarine beds*	*North Yorkshire*
10, 11	*calcareous sandstone from the Millepore bed*	*North Yorkshire*
12	*sandstone from the Ellerbeck bed.*	*North Yorkshire*

All three limestone samples from near the base of the Estuarine beds of Shye were reversed, and these results (Fig. 1) combined with those from a sandstone sample from the Brora Arenaceous Series gave the following statistics:

Declination S 54·4 W, inclination $-66°$, $\alpha = 6·6°$, $\kappa = 33·5$, $N = 14$, where κ is an estimate of the precision, and N represents the number of disks measured. The mean declination and inclination of the individual samples are given in the following table.

TABLE III

Mean declination and inclination of the Scottish Jurassic samples (samples numbers correspond to Fig. 1)

	Declination	Inclination		Declination	Inclination
1	S65E	+78	4	S52W	−64
2	N43W	+78	5	S60W	−53
3	S77W	−78	6	S40W	−46

The intensity of magnetization of the limestone samples ranged between 0·3 and 8×10^{-4} e.m.u./cm^3.

The corresponding pole position is in long. 76° E., lat. 56° N., a point about 350 miles north-east of Omsk; the opposite pole lies in the Pacific just beyond the present northern limit of pack-ice.

Only two of the Skye Liassic limestones were measurable, one was normal the other reversed. The means of the samples (plotted on the Wulff stereographic net) are shown in Fig. 1.

4. *Sedimentary environment.*—The Cretaceous deposits of the Wealden "Lake" (Edmunds (**8**), pp. 19–21), and the Yorkshire and Scottish Estuarine (Deltaic) Series which include some marine horizons are both essentially deltaic deposits laid down in an area with a climate generally considered to be at least warm temperate. This is suggested by the Jurassic coral reefs, and coal, and the flora and saurian fauna of the Weald Cretaceous. The palaeotemperatures obtained from O^{18} determinations by Urey *et al.* (**20**) give a sea temperature up to 10 deg. F greater than at the present day. The source of the sedimentary material, particularly in the Jurassic, lay to the north or north-east, and being obviously from a higher altitude as well as a higher latitude, the climate was presumably not as warm.

Since there is little evidence of reworking, and in the case of current bedded units this possibility can be ruled out, and since the grains are sub-rounded or polished, it seems probable that the hydrated oxide, goethite, is a product of weathering in the source area. Moore and Maynard (**21**) have shown that under deltaic conditions any iron, whether carried as an organic colloid or as a soluble chemical compound, tends to be deposited in a "limonitic" form, that is as microcrystalline goethite with absorbed water (Posnjak and Merwin (**22**)). This precipitated iron oxide hydrate will tend to occur interstitially.

Such pre-depositional weathering under temperate conditions which results ultimately in the formation of goethite-bearing rocks distinguishes such deposits from the magnetically suitable red sediments in which the iron mineral is characteristically preserved in the anhydrous haematite form.

5. *Conclusions.*—The Skye limestones and the Sutherland sandstone, suggest a reversed magnetic field in mid-Jurassic times. The magnetic inclination had much the same value as at the present day, although a pole position near Omsk

is suggested from the declination. No other stable rocks were found, and the majority of samples proved to be too weakly magnetized for measurement under the astatic magnetometer.

From the magnetic point of view the differing, climatically-dependent weathering conditions of the source areas of drab and red beds is probably as important a difference as their respective deltaic and continental depositional histories. Deltaic deposits whose source areas lie in a cold or glacial climate may be palaeomagnetically suitable because of the dominance of mechanical over chemical weathering.

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